01

MGU BCA - 3RD SEM

Quantitative Techniques

Module 1 - Introduction to Statistics

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Complete question bank: every important, repeated and sure-mark question for this module, each with a full-mark model answer. Nothing important left out.

EXAM PATTERN (from real MGU question papers)

EXAM PATTERN (QT): The real MG3CCRBCA200 paper pattern is UNKNOWN - no Honours model paper and no past BCA paper exists online (first exam 22-10-2025 was never published). Closest evidence: MGU B.Com 'Quantitative Techniques for Business' papers (same university, overlapping statistics syllabus) run 80 marks / 3 hours: Part A answer any 10 x 2 marks = 20 (short theory + one-line numericals), Part B answer any 6 x 5 marks = 30 (medium theory + numericals), Part C answer any 2 x 15 marks = 30 (long numericals: combined mean/SD/CV two-firm problem repeated 3x, Bayes theorem). QT is a 4-credit course, so the Honours paper may instead follow a 70-mark / 2-hour format (like DBMS) - this is NOT confirmed. Prepare every module: Part A 2-mark questions span all modules.

TOPIC 1

What is Statistics?

1

Statistics is the science of collecting, organising, presenting, analysing and interpreting numerical data.

Used in two senses: statistical data (numbers and facts) and statistical methods (techniques used on data).

In short: statistics turns raw numbers into decisions.

QUESTION BANK

1 **Define statistics and explain its meaning.**

IMPORTANT

Statistics is the science of collecting, organising, analysing, interpreting and presenting numerical data so that sensible conclusions can be drawn from them.

The word statistics comes from the Latin word 'status' meaning a state, because rulers first collected facts about population, army and revenue of the state.

In simple meaning, statistics is the set of methods used to handle numerical facts. For example, the marks of 500 students in a college are just raw numbers, but when we find the average mark and the highest and lowest marks, we are using statistics.

Statistics does not deal with one or two isolated facts. It always deals with a large collection of facts, because conclusions are drawn from the group as a whole.

The main purpose of statistics is to turn confusing raw data into clear information. It replaces guesswork with a systematic way of studying facts and making decisions.

For example, a shopkeeper who records daily sales for a month and finds which item sold most is using statistics in the simplest form.

In short, statistics is the science of learning from numbers.

2 Explain the two senses in which the word statistics is used.

IMPORTANT

The word statistics is used in two different senses: singular and plural.

In the singular sense, statistics means the science of statistics. It is the body of methods used to collect, analyse and interpret numerical data, just like physics or mathematics is a science.

In the plural sense, statistics means statistical data. Here it refers to the numerical facts themselves, like the population of India, the rainfall in Kerala, or the prices of rice in a market.

For example, the statement 'the statistics of road accidents show a rise' uses statistics in the plural sense, meaning the actual figures.

On the other hand, the statement 'she studied statistics in college' uses statistics in the singular sense, meaning the subject or science.

Both senses are connected: the science of statistics (singular) is applied to statistical data (plural) to draw useful conclusions.

3 'Statistics is the science of averages.' Explain.

IMPORTANT

The statement 'statistics is the science of averages' means that an average is the most common and most useful tool in statistics.

An average represents the whole group of numbers by one single figure. For example, instead of remembering the salaries of 200 workers, one average salary gives the general idea of the group.

Averages make large data easy to understand. The average marks of a class tell us quickly whether the class performed well or badly.

Averages also help in comparison. We can compare the average rainfall of two districts, or the average profit of a company in two years, using a single figure.

Economists and businessmen use averages daily, for example average price, average cost, average production and average demand.

However, the statement is not fully complete. Averages are very important, but statistics also uses other tools like diagrams, correlation and forecasting, so statistics is much more than just averages.

In short, statistics can be called the science of averages because the average is its most widely used product, though not its only one.

TOPIC 2

2

Functions of Statistics

Simplifies complex data into a few meaningful figures like averages.

Makes comparison between two or more data sets easy.

Reveals relationships between factors, like price and demand.

Helps in forecasting the future and framing policies.

QUESTION BANK

4 **Explain the functions of statistics. (2024, 2022)**

REPEATED

One function of statistics is to simplify complex data. Raw data like the monthly sales of 100 shops is too large to understand, but an average or a percentage makes it simple.

Second, statistics helps in comparison. Figures from different years, places or companies can be compared only after they are put into a common form like averages or percentages.

Third, statistics helps in forming policies. A government studies unemployment figures before launching a job scheme, and a company studies sales data before expanding production.

Fourth, statistics helps in forecasting. Past data studied with statistical methods can predict future trends, for example predicting next year's demand from the sales of past five years.

Fifth, statistics helps in measuring relationships. It can show whether two things move together, like whether advertising expenditure and sales are related.

Sixth, statistics helps in testing old ideas. An old belief, like 'demand falls when price rises', can be checked with real data instead of mere opinion.

Finally, statistics enlarges human knowledge. Scientific research in medicine, agriculture and social sciences depends on statistical analysis of experiments.

In short, statistics collects, simplifies, compares and interprets data so that sound decisions and predictions become possible.

5 **Explain the uses of statistics in business and economics.**

IMPORTANT

In business, statistics is used to plan production. A company studies past sales figures to decide how much to produce next month, so that there is neither shortage nor waste.

Statistics helps in studying the market. Before launching a new product, firms collect data on customer preferences, income and buying habits to judge the demand.

It is used in quality control. A factory checks a small sample of its products to judge whether the whole production is of good quality, instead of checking every single piece.

It helps in fixing prices and measuring profit. Costs, sales and profits of different periods or branches are compared using averages and percentages.

In economics, statistics is used to measure the health of the economy. Figures like national income, price index, unemployment rate and growth rate are all statistical products.

It helps in studying trade and prices. A businessman watches the price index and market trends to decide when to buy raw materials and when to sell finished goods.

Governments use statistics for economic planning. The five-year plans and budgets of India are prepared from data on population, agriculture, industry and trade.

In short, statistics is the backbone of business decisions and economic policy, because both run on facts, not on guesses.

TOPIC 3

Limitations of Statistics

3

Studies only numerical facts; feelings and opinions cannot be measured directly.

Studies groups and averages, not individual cases.

Results are true only on an average, never 100% exact.

Statistics can be misused to mislead; it needs expert interpretation.

QUESTION BANK

6 'Statistics can be misused.' Explain with examples. (2018, 2019)

REPEATED

Statistics can be misused means that the same set of figures can be presented in a dishonest way to create a false impression.

One common misuse is choosing a convenient time period. A company that grew from Rs. 10 lakh profit to Rs. 11 lakh in ten years can still say 'profit doubled' if it picks a low-profit base year.

A second misuse is the wrong type of average. If five workers earn Rs. 8,000 each and the owner earns Rs. 5,00,000, the company can proudly say 'average salary Rs. 90,000', which hides the truth.

A third misuse is misleading diagrams. A graph with a very small scale can make a tiny rise in sales look like a huge jump, fooling the viewer at first glance.

A fourth misuse is biased sampling. A company can test only its satisfied customers and then claim '90 percent of customers are happy', while the unhappy ones were never asked.

Another misuse is quoting percentages without the base. A shop can claim 'sales increased 100 percent' when sales went from 2 items to 4 items, which sounds big but is actually small.

The misuse is not the fault of statistics itself. It is the fault of dishonest users, and the remedy is to check the source, method and base of every figure before believing it.

7 Explain the limitations of statistics.

IMPORTANT

Statistics is a powerful tool, but it has clear limitations that every student must know.

First, statistics deals only with numerical facts, so qualities like honesty, beauty or intelligence cannot be studied directly unless converted into numbers.

Second, statistics studies groups and averages, never individuals. An average salary of Rs. 30,000 tells us nothing about what any single person earns.

Third, statistical results are true only on an average. They describe the general tendency of the group, not every single case, so they are never 100 percent exact.

Fourth, statistics can be deliberately misused. The same data can be presented in misleading ways, for example by choosing a convenient time period or by drawing a graph with an exaggerated scale.

Fifth, statistics does not reveal the full story behind the numbers. Two classes can have the same average marks but very different spreads of marks.

Finally, statistical conclusions need expert interpretation. Wrong methods or careless reading

lead to wrong decisions.

In short, statistics is a servant, not a master: it guides decisions but cannot replace judgement and common sense.

TOPIC 4

4

Population and Sample

Population is the complete set of all items under study. Example: all BCA students in Kerala.

Sample is a small part of the population selected for study. Example: 100 students picked at random.

Studying a sample saves time and money; results are extended to the population.

QUESTION BANK

8 **Distinguish between population and sample.**

IMPORTANT

Population is the complete collection of all items or persons we want to study. A sample is only a part of the population that is actually observed.

For example, if we want to study the height of all BCA students in Kerala, all those students are the population, while 200 students selected from them are the sample.

A population is usually very large and hard to study fully, while a sample is small and manageable, so studying a sample saves time and money.

Information collected from a population is always complete and accurate, but it is expensive and slow. Information from a sample is quick and cheap, but it carries a small chance of error.

Results from a population are exact facts, while results from a sample are only estimates of the population, though a good sample gives a close estimate.

In short, the population is the whole field of study and the sample is a small representative part of it.

9 **Define population, sample, parameter and statistic.**

IMPORTANT

Population is the total set of all units under study. For example, all the students of a college form the population if we are studying that college.

Sample is a small part of the population selected for actual observation. For example, 50 students chosen from the college are a sample.

Parameter is a numerical value that describes the population. For example, the average height of ALL students of the college is a parameter. It is usually denoted by Greek letters like mu.

Statistic is a numerical value that describes the sample. For example, the average height of the 50 selected students is a statistic. It is usually denoted by English letters like x-bar.

A parameter is fixed but usually unknown, because studying the whole population is hard. A statistic is known from the sample and is used to estimate the parameter.

In short, a parameter belongs to the population and a statistic belongs to the sample, and statistics uses the sample statistic to guess the population parameter.

TOPIC 5

Data: Types, Collection, Questionnaire

5

Qualitative data describes qualities (gender, colour); quantitative data is numerical (marks, height).

Primary data is collected fresh by the researcher; secondary data already exists (books, reports, websites).

Primary methods: direct interview, questionnaire, observation, telephonic survey.

A good questionnaire is short, with simple clear questions in logical order and no leading questions.

QUESTION BANK

10 **Distinguish between primary and secondary data.**

IMPORTANT

Primary data is data collected fresh for the first time for a specific purpose. Secondary data is data already collected by someone else for some other purpose and reused by us.

For example, interviewing 100 customers about a new product gives primary data, while using the government census report of 2011 gives secondary data.

Primary data is original and fits the present purpose exactly, but collecting it takes time and money. Secondary data is quick and cheap, but it may not fit our exact need.

Primary data is more reliable for our purpose because we control how it is collected. Secondary

data may contain errors or outdated facts, so it must be checked before use.

Primary data is always collected by the investigator himself through surveys or experiments. Secondary data comes from published reports, newspapers, journals and websites.

In short, primary data is first-hand and tailor-made, while secondary data is second-hand and ready-made.

11 What are the methods of collecting primary data?

IMPORTANT

The main methods of collecting primary data are direct personal investigation, indirect oral investigation, questionnaires, schedules, and direct observation.

Direct personal investigation means the investigator himself meets the persons and collects answers. It gives accurate data, but it is costly and suitable only for a small area. For example, a researcher personally interviewing 30 shopkeepers in a town.

Indirect oral investigation means the investigator collects information from other persons who know about the subject, called witnesses. It is used when direct contact is not possible. For example, asking employers about workers' habits.

The questionnaire method means printed questions are sent by post or email to the informants, who fill and return them. It covers a wide area cheaply, but it works only with educated respondents.

The schedule method is similar, but the investigator himself visits the informant, asks the questions and fills the form. It gives better response than the postal questionnaire.

Direct observation means the investigator watches and records without asking questions. For example, counting the number of customers entering a shop per hour.

Telephone and online surveys are modern methods, quick and cheap, but they reach only people with phones or internet access.

12 What are the essentials of a good questionnaire?

IMPORTANT

A good questionnaire must have a clear purpose. Every question should be related to the aim of the study, and useless questions must be avoided.

The questions must be simple and easy to understand. Technical words and confusing language should not be used, because the respondent may answer wrongly.

Questions should be short and few in number. A long questionnaire bores the respondent and increases the chance that he will not reply.

Personal and embarrassing questions should be avoided, or asked very carefully, because the respondent may refuse to answer or give false answers.

Questions should be arranged in a logical order, starting with easy general questions and moving to specific ones, so the respondent feels comfortable.

Leading questions must be avoided. A question like 'Do you not agree our product is best?' pushes the answer, so neutral wording is needed.

Enough space must be given for answers, and instructions at the top must clearly explain how to fill the form.

A good questionnaire should be tested on a few people first, called a pilot survey, so that mistakes can be corrected before the real survey.

In short, a good questionnaire is simple, short, clear, unbiased and easy to fill.

13 Distinguish between qualitative and quantitative data with examples.

IMPORTANT

Qualitative data describes qualities or attributes that cannot be measured in numbers. Quantitative data consists of facts that can be measured and expressed in numbers.

Examples of qualitative data are gender, religion, colour, beauty, honesty and marital status. We can only classify them into groups, not measure them.

Examples of quantitative data are height, weight, age, income, marks and temperature. These can be counted or measured exactly.

Qualitative data is studied by counting how many units fall in each category, for example how many students are male and how many are female.

Quantitative data is studied with averages and other numerical tools, for example the average income of workers or the average marks of a class.

Sometimes qualitative facts are converted into numbers, like rating beauty from 1 to 10, but the original fact is still a quality, not a true measurement.

In short, qualitative data tells us the kind of thing, while quantitative data tells us how much of it.

14 Draft a questionnaire for collecting socio-demographic details of women workers in a factory. (2024)

IMPORTANT

A questionnaire is a printed set of questions used to collect primary data directly from respondents.

For socio-demographic details of women workers, it must be simple, respectful and in the local language.

It should start with a short introduction stating the survey purpose and promising that answers stay confidential.

First come personal details: age group, marital status, number of children and education level.

Next come job details: department, designation, years of service, monthly income and shift timing.

Then come family details: type of house, total family income, distance from the factory and mode of transport.

Questions should be short and mostly tick-box type, so that even busy or less-educated workers can answer fast.

Avoid embarrassing or leading questions; never ask anything that hurts dignity or privacy.

Keep the form short, about 15 to 20 questions, because a long form tires workers and gives careless answers.

Example items: 'Age: below 25 / 25-35 / 35-45 / above 45' and 'Monthly income: below 10000 / 10000-20000 / above 20000'.

End with a thank-you line for the worker's time.

In short, this questionnaire is short, simple and respectful, covering personal, job and family details.

TOPIC 6

6

Census, Sampling and Types of Sampling

Census collects data from every unit; sampling studies only a selected part.

Simple random sampling: every unit has an equal chance, like a lottery.

Stratified sampling: split into groups first, then sample randomly from each group.

Systematic sampling: pick every kth unit, e.g. every 10th customer.

Cluster sampling: pick whole natural groups at random, like villages or classes.

Convenience and judgement sampling are quick but biased non-random methods.

QUESTION BANK

15 **Explain the various types of sampling with examples.**

IMPORTANT

The main types of sampling are simple random sampling, stratified sampling, systematic sampling and cluster sampling, plus judgement sampling.

Simple random sampling means every unit has an equal chance of being selected. For example, writing the names of 500 students on slips, mixing them, and drawing 50 slips. It is the fairest method.

Stratified sampling means the population is first divided into groups called strata, like boys and girls, and then a random sample is taken from each group. It ensures every group is represented.

Systematic sampling means selecting every kth unit after a random start. For example, from a list of 1,000 customers, selecting every 10th customer. It is simple and quick.

Cluster sampling means the population is divided into clusters, like villages or classes, and a few whole clusters are selected at random. All units in the chosen clusters are studied.

Judgement sampling means the investigator selects the units using his own judgement, like choosing 20 shops he considers typical. It is quick but can be biased.

Probability methods like random, stratified, systematic and cluster sampling give every unit a known chance of selection, while judgement sampling is non-probability and depends on the investigator.

16 **Distinguish between census and sample survey.**

IMPORTANT

A census survey studies every single unit of the population. A sample survey studies only a small selected part of it.

For example, the Indian census counts every person in the country, while an opinion poll interviews only a few thousand voters.

A census gives complete and exact information, but it takes a long time and costs a lot of money, so it is done rarely, like once in ten years.

A sample survey is quick and cheap, so it can be done often, but its results are only estimates and carry a small sampling error.

A census needs a huge staff and organisation. A sample survey needs fewer people and simpler organisation.

A census is suitable when the population is small or when full accuracy is needed. A sample survey is suitable when the population is large and quick results are needed.

In short, a census studies all and a sample studies a few, and the choice depends on time, money and the accuracy needed.

17 What is stratified sampling? Explain with an example.

IMPORTANT

Stratified sampling is a method in which the population is first divided into homogeneous groups called strata, and then a random sample is taken from each stratum.

The strata are formed on the basis of some common characteristic, like age, income, gender or class, so that units inside one stratum are similar.

Sampling from every stratum ensures that all types of units are represented in the sample, which a simple random sample may miss by chance.

For example, a college with 600 boys and 400 girls wants a sample of 100 students. It can take 60 boys at random from the boys' list and 40 girls at random from the girls' list.

The number taken from each stratum is usually in proportion to the size of the stratum, which is called proportionate stratified sampling.

This method gives more accurate results than simple random sampling when the population is mixed, because each group contributes its fair share to the sample.

18 Discuss on multi-stage sampling. (2024)

IMPORTANT

Multi-stage sampling is a probability sampling method in which the sample is selected in two or more stages.

At the first stage, large primary units are selected at random; at later stages, smaller units are picked from within them.

For example, to survey college students in Kerala: first select 5 districts at random, then 10 colleges from each district, then 20 students from each college.

It is very useful when the population is spread over a large area, because the investigator need not visit every corner.

It saves time and money compared to taking a simple random sample of the whole scattered population.

Its limitation is that sampling errors can add up at each stage, so the final sample may be less accurate.

In short, multi-stage sampling picks the sample step by step, from large groups down to the final units.

19 Discuss on systematic sampling. (2022)

IMPORTANT

Systematic sampling is a probability sampling method in which every kth unit of the population is selected after a random start.

First the population list is arranged in order; then the sampling interval is fixed as k = N/n, where N is population size and n is sample size.

A random number between 1 and k is chosen as the starting point, and then every kth unit is picked.

For example, to select 100 students from 1000: k = 1000/100 = 10; with a random start of 7, select roll numbers 7, 17, 27, 37 and so on.

Its merit is simplicity: only the first unit is chosen at random, the rest follow automatically, and the sample spreads evenly over the list.

Its limitation: if the list has a hidden repeating pattern that matches k, the sample becomes biased.

In short, systematic sampling selects every kth unit after one random start.

20 Explain the non-probability sampling techniques. (2022)

IMPORTANT

In non-probability sampling the units are not selected by chance; the investigator uses judgement or convenience.

The main types are judgement (purposive) sampling, convenience sampling, quota sampling and snowball sampling.

In judgement sampling the investigator deliberately picks units he considers typical, like selecting model farmers for an agriculture study.

In convenience sampling whoever is easily available is selected, like interviewing shoppers in one particular mall.

In quota sampling fixed quotas are set for each group, for example 50 men and 50 women, and the investigator fills them as he likes.

In snowball sampling one respondent leads to the next; it suits hidden populations whose members are hard to find directly.

The merit is speed and low cost; the limitation is that results cannot be safely generalised to the

whole population.

In short, non-probability sampling is quick and cheap but not fully representative.

21 What are the advantages of sample survey? (2018)

IMPORTANT

A sample survey studies only a part of the population instead of every single unit.

Its first advantage is lower cost: studying 500 people costs far less than studying five lakh people.

Second, it saves time, so results come quickly, which matters for urgent decisions like election forecasts.

Third, the data is often more accurate, because a small team of investigators can be trained well and supervised closely.

Fourth, it is the only possible method when testing destroys the unit, like testing the life of electric bulbs.

Fifth, it needs fewer investigators and less organisation than a full census.

In short, a sample survey is cheaper, faster and often more reliable than a census.

22 What are the principal steps involved in the planning and execution of a sample survey? (2018)

IMPORTANT

A sample survey must be planned and executed in a clear sequence of steps.

First, define the objective: exactly what information is needed and for what purpose.

Second, define the population and prepare the sampling frame, the complete list of all units.

Third, choose the sampling method (simple random, stratified, systematic and so on) and fix the sample size.

Fourth, design the questionnaire or schedule and test it on a small pilot group.

Fifth, recruit and train the investigators, and fix the fieldwork timetable and budget.

Sixth, collect the data in the field with proper supervision and surprise checks.

Seventh, edit, classify and tabulate the collected data, removing errors and inconsistencies.

Finally, analyse the data, write the report and state the conclusions along with their limitations.

In short, a sample survey moves step by step from objective to frame to method to fieldwork to report.

TOPIC 7

7

Classification and Tabulation

Classification groups data by common features: geographical, chronological, qualitative or quantitative.

Tabulation presents classified data neatly in rows and columns.

Parts of a table: title, stub (row headings), caption (column headings), body (figures), footnote, source.

QUESTION BANK

23 **What is tabulation? Explain the parts of a statistical table. (2024, 2018, 2019)**

SURE MARK

Tabulation is the systematic arrangement of data in rows and columns in the form of a table, so that the data becomes easy to read and compare.

The first part of a table is the table number, which identifies the table, like Table 1 or Table 2.

The second part is the title, written at the top, which clearly states what the table is about, like 'Marks of BCA students, 2026'.

The third part is the stub, which is the left-hand column containing the row headings, like the names of classes or years.

The fourth part is the caption, which contains the column headings at the top, like 'Boys', 'Girls' and 'Total'.

The fifth part is the body of the table, which contains the actual figures arranged in rows and columns.

The sixth part is the head note, written just below the title in brackets, giving the unit of measurement, like '(in thousands)' or '(in Rs.)'.

The last part is the foot note, written at the bottom, which explains anything unusual about the data or its source.

A good table is simple, clear, well-titled and complete, so that anyone can understand the data at a glance.

24 What is classification? Explain its types.

IMPORTANT

Classification is the process of arranging data into groups or classes according to some common characteristic, so that similar facts come together.

The first type is geographical classification, where data is grouped by place. For example, population of India classified by states like Kerala, Tamil Nadu and Karnataka.

The second type is chronological classification, where data is grouped by time. For example, the sales of a company classified year-wise from 2020 to 2026.

The third type is qualitative classification, where data is grouped by attributes that cannot be measured. For example, students classified as male or female, or as pass or fail.

The fourth type is quantitative classification, where data is grouped by numerical values. For example, students grouped by marks like 0-10, 10-20 and 20-30.

Qualitative classification can be simple, with two classes like literate and illiterate, or manifold, with many classes like Hindu, Muslim and Christian.

Quantitative classification produces a frequency distribution, where each class shows how many units fall in it, like 25 students in the 60-70 marks class.

In short, classification organises raw data into meaningful groups, and it is the first step before tabulation.

25 Write a note on source note. (2024)

IMPORTANT

A source note is a short line written below a statistical table stating where the data was taken from.

It is usually placed at the bottom left of the table, below the footnotes.

Its purpose is to give credit to the original source and to let the reader verify the figures if needed.

For example: 'Source: Census of India, 2011' or 'Source: Annual Report of XYZ Company, 2023-24'.

Without a source note the reader cannot judge whether the figures are trustworthy.

It is one of the essential parts of a good table, along with the title, captions, stubs, body and footnotes.

In short, the source note tells the reader the origin of the table's figures.

26 **Form a frequency distribution from the following data by the exclusive method, taking 10 as the magnitude of class intervals: 10, 17, 15, 22, 11, 16, 19, 24, 29, 18, 25, 26, 32, 14, 17, 20, 23, 27, 30, 12. (2019)**

IMPORTANT

In the exclusive method the upper limit of a class is excluded and counted in the next class, so 10-20 means 10 up to but below 20.

First arrange the 20 values in order: 10, 11, 12, 14, 15, 16, 17, 17, 18, 19, 20, 22, 23, 24, 25, 26, 27, 29, 30, 32.

With class magnitude 10, the classes are 10-20, 20-30 and 30-40.

Class 10-20 takes 10, 11, 12, 14, 15, 16, 17, 17, 18, 19, giving frequency 10.

Class 20-30 takes 20, 22, 23, 24, 25, 26, 27, 29, giving frequency 8.

Class 30-40 takes 30, 32, giving frequency 2.

Check the total: 10 + 8 + 2 = 20, which equals the number of given values.

Answer: the frequency distribution is 10-20: 10, 20-30: 8, 30-40: 2.

27 **Form a frequency distribution from the following data by the inclusive method, taking 4 as the magnitude of class intervals: 10, 17, 15, 22, 11, 16, 19, 24, 29, 18, 25, 26, 32, 14, 17, 20, 23, 27, 30, 12, 15, 18, 24, 36, 18, 15, 21, 28, 33, 38, 34, 13, 10, 16, 20, 22, 29, 19, 23, 31. (2022)**

IMPORTANT

In the inclusive method both limits of a class are included, so 10-13 means the values 10, 11, 12 and 13.

With magnitude 4 starting from 10, the classes are 10-13, 14-17, 18-21, 22-25, 26-29, 30-33, 34-37, 38-41.

Counting the 40 values into these classes: 10-13 gets 5 values (10, 10, 11, 12, 13).

14-17 gets 8 values (14, 15, 15, 15, 16, 16, 17, 17).

18-21 gets 8 values (18, 18, 18, 19, 19, 20, 20, 21).

22-25 gets 7 values (22, 22, 23, 23, 24, 24, 25).

26-29 gets 5 values (26, 27, 28, 29, 29).

30-33 gets 4 values (30, 31, 32, 33).

34-37 gets 2 values (34, 36) and 38-41 gets 1 value (38).

Check: 5+8+8+7+5+4+2+1 = 40, the total number of values.

Answer: frequencies are 5, 8, 8, 7, 5, 4, 2, 1 for the eight classes in order.

28 **Write a short note on cross tabulation. (2022)**

IMPORTANT

Cross tabulation is a table that presents the joint distribution of two characteristics at the same time.

The rows show one characteristic and the columns show the other, and each cell gives the count for that combination.

For example, a table of 200 college students with rows Male/Female and columns Hosteller/Day scholar shows male hostellers, female hostellers and so on.

It is very useful for studying the relationship between two qualitative variables.

From a cross table we can compute row percentages and column percentages to compare groups easily.

Its limitation is that with more than two variables the table becomes large and difficult to read.

In short, cross tabulation classifies data by two characteristics together in a single table.

29 **The following are the amounts of income tax paid by 600 businessmen in year 2010: Income tax Rs (more than) 500: 600, 1000: 550, 1500: 425, 2000: 275, 2500: 100, 3000: 25. Find the number of businessmen who paid more than Rs. 1,200 but not more than Rs. 2,400 as income tax. (2022)**

IMPORTANT

This is a 'more than' cumulative frequency table; first convert it into ordinary class frequencies.

500-1000: 600-550 = 50; 1000-1500: 550-425 = 125; 1500-2000: 425-275 = 150.

2000-2500: 275-100 = 175; 2500-3000: 100-25 = 75; above 3000: 25.

We need businessmen paying more than 1200 but not more than 2400.

From class 1000-1500 (125 persons), the part above 1200 is (1500-1200)/500 = 0.6, giving 0.6 x 125 = 75.

The full class 1500-2000 (150 persons) lies inside the range, so take all 150.

From class 2000-2500 (175 persons), the part up to 2400 is (2400-2000)/500 = 0.8, giving 0.8 x 175 = 140.

Total = 75 + 150 + 140 = 365.

Answer: 365 businessmen paid more than Rs. 1,200 but not more than Rs. 2,400.

TOPIC 8

8

Diagrams and Graphs

Bar diagrams compare values with bars: simple, multiple or subdivided.

Pie diagram: a circle split into sectors; sector angle = 360 x (component value / total).

Histogram: bars for continuous frequency data with no gaps between bars.

Frequency polygon and curve: line forms joining the midpoints of histogram tops.

Ogives: cumulative frequency curves (less-than and more-than), used to find the median graphically.

QUESTION BANK

30 **How do you construct a pie diagram? Explain with the formula.**

IMPORTANT

A pie diagram is a circle divided into sectors, where each sector represents one part of the data and the size of the sector is proportional to the value it represents.

The first step is to find the total of all the values in the data.

The second step is to calculate the angle for each item using the formula: angle = (value of the item / total value) x 360 degrees, because a full circle has 360 degrees.

For example, if a family spends Rs. 900 on food, Rs. 600 on rent and Rs. 300 on others out of Rs. 1,800 total, the angles are: food = (900/1800) x 360 = 180 degrees, rent = (600/1800) x 360 = 120 degrees, others = (300/1800) x 360 = 60 degrees.

The third step is to draw a circle and divide it into sectors with these angles using a protractor, starting from a fixed point and moving clockwise.

Each sector is labelled or shaded differently, with a clear legend showing which sector stands for which item.

Pie diagrams are best when the data is divided into a few parts and we want to show the share of each part in the whole, like budget shares or market shares.

A pie diagram makes comparison of parts easy at a glance, because a bigger slice clearly means a bigger share.

31 Write a short note on ogives.

IMPORTANT

An ogive is a graph of the cumulative frequency distribution, drawn as a smooth freehand curve. It is also called a cumulative frequency curve.

To draw it, the upper limits of the class intervals are marked on the x-axis and the cumulative frequencies on the y-axis, and the points are joined by a smooth curve.

There are two types of ogives. The 'less than' ogive is drawn using less-than cumulative frequencies, and its curve rises from left to right.

The 'more than' ogive is drawn using more-than cumulative frequencies, and its curve falls from left to right.

The main use of ogives is to find the median and other partition values like quartiles. The point where the two ogives cross gives the median of the data.

For example, to find the median marks of a class, we draw both ogives and read the x-value at their intersection point.

Ogives are also useful for comparing two distributions and for finding how many observations lie above or below a given value.

In short, an ogive turns a frequency table into a rising or falling curve that helps locate the median quickly.

32 Distinguish between a histogram and a bar diagram.

IMPORTANT

A histogram is drawn for continuous frequency data, while a bar diagram is drawn for discrete or categorical data.

In a histogram, the bars are joined together with no gaps, because the classes are continuous, like 0-10, 10-20. In a bar diagram, the bars are separated by gaps, because the categories are distinct, like years or products.

In a histogram, the x-axis shows class boundaries and the area of each bar represents the frequency. In a bar diagram, only the height of the bar represents the value.

A histogram can have unequal class widths, in which case the heights are adjusted so that the area stays proportional. A bar diagram always has bars of equal width.

A histogram cannot be drawn for qualitative data like gender or religion, but a bar diagram can easily show such categories.

For example, the marks of students in class intervals are shown by a histogram, while the sales of five different products are shown by a bar diagram.

In short, a histogram shows a continuous distribution with touching bars, while a bar diagram compares separate categories with spaced bars.

33 What is a frequency polygon? How is it drawn?

IMPORTANT

A frequency polygon is a line graph obtained by joining the mid-points of the tops of the rectangles of a histogram with straight lines.

To draw it, first the mid-points of each class are marked on the x-axis and the frequencies on the y-axis, and the points are plotted.

The plotted points are then joined by straight lines, forming a polygon, which gives the graph its name.

Two extra classes with zero frequency are added, one before the first class and one after the last class, so that the polygon touches the x-axis at both ends and looks complete.

A frequency polygon can also be drawn directly without first drawing a histogram, by plotting class mid-points against frequencies.

For example, for the marks classes 0-10, 10-20 and 20-30 with frequencies 5, 12 and 8, we plot the points (5, 5), (15, 12) and (25, 8) and join them.

Frequency polygons are useful for comparing two or more distributions on the same graph, because several polygons can be drawn together without confusion.

In short, a frequency polygon is a multi-sided line graph that shows the shape of a frequency distribution.